

**DESIGN AND IMPLEMENTATION OF A SMART ATTENDANCE
SYSTEM USING ACOUSTIC AND BLUETOOTH LOW ENERGY
PROXIMITY VERIFICATION**

BY

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21/52HP071

A FINAL YEAR PROJECT SUBMITTED TO THE DEPARTMENT OF
TELECOMMUNICATION SCIENCE,
FACULTY OF COMMUNICATION AND INFORMATION SCIENCES,
UNIVERSITY OF ILORIN, ILORIN, NIGERIA

IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE AWARD OF THE
DEGREE OF BACHELOR OF SCIENCE (B.Sc.) IN TELECOMMUNICATION SCIENCE

JULY 2026

DECLARATION

I, Joshua Olugbemi Iyanuoluwa, with matriculation number 21/52HP071, declare that this project titled “Design and Implementation of a Smart Attendance System Using Acoustic and Bluetooth Low Energy Proximity Verification” is my original work, carried out in the Department of Telecommunication Science, University of Ilorin. All ideas, findings, and words obtained from other authors have been acknowledged through appropriate citation and referencing. This work has not been submitted, either wholly or partly, to this or any other institution for the award of a degree or qualification.

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Signature and Date

CERTIFICATION

This is to certify that the project titled “Design and Implementation of a Smart Attendance System Using Acoustic and Bluetooth Low Energy Proximity Verification” was carried out by Joshua Olugbemi Iyanuoluwa (21/52HP071) in the Department of Telecommunication Science, University of Ilorin, under approved academic supervision. The project has been examined and found to meet the requirements for the award of the Bachelor of Science (B.Sc.) degree in Telecommunication Science.

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DEDICATION

This work is dedicated to God Almighty, whose grace and strength sustained me throughout the course of this study; to the cherished memory of my late father; and to my family, whose love, sacrifice, and encouragement have remained a constant source of support.

ACKNOWLEDGEMENTS

I give all glory to God Almighty for the wisdom, strength, and opportunity to complete this project.

I sincerely appreciate my project supervisor for the guidance, constructive observations, and academic direction provided during the development of this work. I am also grateful to the academic and technical staff of the Department of Telecommunication Science, University of Ilorin, for the knowledge and support that contributed to my training.

My profound gratitude goes to my family for their patience, sacrifices, prayers, and unwavering encouragement. I equally thank my colleagues and friends whose discussions, practical assistance, and willingness to participate in testing helped the project progress from an idea to a working implementation.

ABSTRACT

This study designed and implemented a mobile attendance system that uses acoustic beacons and Bluetooth Low Energy (BLE) proximity evidence to support classroom attendance. A Flutter application provides separate lecturer and student workflows, while native Kotlin components perform acoustic transmission and decoding, BLE advertising and scanning, and foreground broadcast management. A Django REST API deployed on Render validates authenticated submissions and stores accepted records in a relational database. The validation process checks the selected session, signal format and age, registered beacon-room association, received signal strength policy, proof digest, student identity, device binding, duplicate attendance, and student-scoped replay. Development followed an iterative Agile approach. Verification comprised 15 backend API tests, four Flutter tests, Android compilation, and repeated small-scale physical testing with four Android phones and a DX-CP27 Mini beacon. The deterministic software checks passed. Acoustic decoding was normally observed at approximately 1–30 cm and slightly beyond 50 cm in the best quiet, high-volume condition. Lecturer-device BLE was detected by the three student phones at 1–5 m, 5–10 m, and above 10 m under the tested conditions, including some indoor obstructions. The room beacon was also detected and resolved through its registered classroom. These results show that BLE is the practical primary path in the current prototype, whereas acoustic proof provides supplementary very-short-range evidence. The evaluation was limited by estimated distances, the absence of fixed trial counts, four phones, and no full-class or adjacent-room test; consequently, the findings establish prototype feasibility rather than institution-wide performance.

Keywords: smart attendance, acoustic beacon, Bluetooth Low Energy, proximity verification, room beacon, device binding

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Attendance records are used in universities to document participation in scheduled academic activities and to support course administration. In many classrooms, however, attendance is still recorded through roll call, signature sheets, or paper registers. These methods are easy to understand, but they consume lecture time, require manual compilation, and are difficult to audit. They also permit proxy attendance when one student answers or signs for an absent colleague.

Digital attendance systems have been developed to reduce these weaknesses. The technologies reported in the literature include radio-frequency identification (RFID), near-field communication (NFC), quick-response (QR) codes, biometrics, geofencing, Bluetooth, and mobile applications (Rashid, 2024). Such systems can improve the speed of attendance capture and simplify reporting, but digitisation alone does not establish physical presence. For example, a card may be exchanged, a static QR code may be forwarded, or valid account credentials may be used on behalf of another student. Miao et al. (2020) addressed attendance fraud within an RFID system, while Nwabuwe et al. (2023) combined dynamic QR codes, geofencing, and device identity to reduce several forms of proxy attendance. These studies show that identity, time, location or proximity, and replay controls must be considered together.

The widespread availability of smartphones creates an opportunity to perform attendance verification with hardware already carried by lecturers and students. A typical Android smartphone provides a loudspeaker, microphone, Bluetooth radio, network interface, local application storage, and runtime security controls. These facilities can support short-range signal transmission, proximity sensing, authentication, and communication with a

remote validation service. The project is therefore relevant to Telecommunication Science because it applies signal generation, reception, wireless propagation, data encoding, proximity estimation, and networked service delivery to an educational problem.

Bluetooth Low Energy (BLE) is suited to nearby-device discovery because an advertiser can transmit small packets that nearby scanners receive without conventional pairing. Earlier attendance implementations used Bluetooth Smart or BLE beacons to reduce manual attendance effort (Azmi et al., 2018; Lodha et al., 2015; Noguchi et al., 2015). Puckdeevongs et al. (2020) further demonstrated a BLE classroom attendance and positioning framework. BLE signal strength is nevertheless affected by distance, antenna orientation, obstacles, phone hardware, interference, and indoor multipath. Ramirez et al. (2021) found that RSSI-based estimates varied with measurement geometry and environmental complexity. RSSI is consequently treated in this project as approximate proximity evidence rather than an exact distance measurement.

Acoustic communication provides a second proximity channel. Getreuer et al. (2018) demonstrated that commodity smartphone speakers and microphones could exchange short tokens in a near-ultrasonic band. Jia et al. (2022) also used smartphone near-ultrasonic signals for proximity measurement. These studies establish the feasibility of acoustic signalling with consumer devices, but they do not imply identical performance on every handset or in every room. Speaker response, microphone sensitivity, background noise, frequency attenuation, and placement all influence reception. The acoustic channel in this project is therefore used as a supplementary, very-short-range proof of copresence rather than as the principal full-room mechanism.

The system developed in this study combines a Flutter Android application with a Django REST backend. A lecturer creates a class session, selects a registered room, opens attendance, and starts a foreground broadcast service. The service rotates a short-lived

acoustic token and lecturer-device BLE nonce while attendance remains open. A student scans for valid evidence and submits a proof to the hosted backend. The backend accepts either a valid acoustic proof, a current lecturer-device BLE proof, or evidence from a registered BLE beacon assigned to the session room.

A fixed room beacon, such as the DX-CP27 used during development, broadcasts a stable identifier rather than a newly generated session nonce. It is therefore not accepted by identity alone. The backend confirms that the beacon is registered and active, belongs to the room selected for the session, satisfies the configured minimum RSSI rule, and corresponds to an attendance session that is currently open in that room. This room-aware rule allows lecturer-device BLE and fixed beacon coverage to coexist without treating every detectable beacon as valid attendance evidence.

Attendance acceptance is decided by the backend rather than by the scan interface alone. The validation process checks authenticated identity, session status, student and device association, proof format, cryptographic digest, signal freshness where applicable, beacon-room association, RSSI policy, duplicate attendance, and replay use. The design responds to evidence that a static BLE beacon signal can be imitated or forwarded (Kim et al., 2018). It does not claim to eliminate every form of attendance fraud; instead, it combines independent controls to make casual proxy attendance and stale-proof reuse more difficult while retaining a fast classroom workflow.

1.2 Statement of the Problem

Manual attendance in higher institutions consumes time that could be used for teaching and produces records that require additional processing. Passing a register through a large class or calling every name may interrupt the lecture, yet neither method reliably prevents one student from responding or signing for another. Paper records can also be misplaced, altered, damaged, or compiled incorrectly.

Some digital alternatives solve the record-keeping problem without adequately solving the presence problem. A static form or QR code can be shared outside the classroom. RFID and NFC systems require tags or readers and may still authenticate the object rather than the person carrying it. Biometric systems strengthen identity verification but introduce equipment, privacy, environmental, and throughput concerns. GPS-based methods can support geofencing but are not consistently precise within indoor rooms. BLE-based attendance offers practical coverage, but a reusable beacon identifier can be copied or relayed if it is accepted without session and server controls (Kim et al., 2018).

The problem addressed by this study is therefore the absence of a practical smartphone-based attendance process that combines rapid capture with evidence of classroom proximity and server-side fraud controls. The proposed solution must identify the student and active class session, accept evidence only through an authorised acoustic or BLE path, reject duplicate and stale submissions, distinguish registered room beacons from unrelated BLE devices, and maintain records that lecturers can review and export. It must also reflect the measured limitations of the prototype: the present acoustic implementation is reliable only at very short range, whereas BLE provides the main useful classroom coverage.

1.3 Aim and Objectives of the Study

The aim of this study is to design and implement a smart attendance system that uses acoustic and Bluetooth Low Energy proximity verification with server-side validation for classroom attendance management.

The specific objectives are to:

1. design a mobile attendance system through which lecturers can create class sessions, select registered rooms, open attendance, and manage attendance records;

2. implement acoustic and BLE proximity verification, using acoustic proof as very-short-range copresence evidence and BLE proof from either a lecturer-device advertisement or a registered room beacon;
3. develop a backend validation process that verifies session identity, proof integrity and freshness, beacon-room association, RSSI policy, duplicate prevention, replay protection, and student device binding;
4. provide lecturer and student interfaces for signal broadcasting, scanning, proof submission, attendance history, session reports, profile information, permission guidance, and CSV export; and
5. evaluate the implemented system on Android devices under different distance, noise, obstacle, permission, hosting, and beacon conditions.

1.4 Research Questions

The study addresses the following questions:

1. How can acoustic beaconing and BLE advertisements be used to support smartphone-based classroom attendance?
2. How can lecturer-device BLE and registered room beacons coexist without accepting evidence from an unrelated room?
3. Which server-side controls can reduce duplicate attendance, stale-proof reuse, replay, and casual account sharing?
4. What practical limitations affect acoustic and BLE reception on the tested Android devices?
5. How effectively does the implemented workflow support session creation, scanning, proof submission, reporting, and export during small-scale testing?

1.5 Scope of the Study

The study covers the design, implementation, and small-scale evaluation of an Android smart attendance system. Its software components are a Flutter mobile application, native Android acoustic and BLE modules, and a Django REST backend hosted on Render with a PostgreSQL database. SQLite remains available for local development. The application provides separate lecturer and student workflows, while the Django administration interface supports controlled management and inspection of users, sessions, beacons, proofs, and replay records.

The lecturer workflow includes authentication, session creation, room selection, attendance opening and closure, background acoustic and BLE broadcasting, live-session management, session-specific validation reports, search, and CSV export. The student workflow includes authentication, device registration, permission checks, concurrent acoustic and BLE scanning, proof review, one-time submission, attendance history, and profile information.

Three proof sources are within scope:

1. a short-lived acoustic token transmitted by the lecturer phone;
2. a short-lived BLE nonce advertised by the lecturer phone; and
3. a static registered room-beacon identifier validated against an open session, room assignment, and RSSI rule.

The implementation focuses on Android because the acoustic decoder, BLE advertiser, foreground service, wake lock, and runtime permissions use Android-specific facilities. An iOS version is outside the completed implementation. The active attendance paths are limited to acoustic and BLE evidence.

The system verifies proximity evidence; it does not claim precise indoor positioning or conclusive biometric identity. BLE RSSI is not converted into an exact distance, and the

application-generated device identifier is a device-trust control rather than hardware-backed attestation. Institutional-scale load testing, multi-room field trials, formal penetration testing, and accessibility studies are beyond the completed small-scale evaluation.

1.6 Significance of the Study

The study demonstrates how telecommunication and mobile-computing principles can be applied to an administrative problem in higher education. It combines acoustic signal encoding and decoding, BLE advertising and scanning, signal-strength policy, mobile networking, a hosted validation service, and database reporting in one working system.

For lecturers, the system reduces the steps required to collect and compile attendance and produces a report tied to the selected class session. For students, attendance can be submitted from a personal Android device without paper, a contact-based biometric reader, or conventional Bluetooth pairing. For administrators, the backend provides auditable records, registered room-beacon management, duplicate controls, device-binding information, and controlled device-reset actions.

The project also provides evidence for choosing between the two signal paths. Its four-device evaluation shows that the present acoustic design is useful only at close range, while BLE is the stronger operational mechanism. This result is important because it avoids presenting a technically novel acoustic feature as more reliable than the measured evidence supports. The fixed-beacon design offers a route to improved room coverage, while the room and session checks reduce the risk of accepting every beacon detected nearby.

Finally, the implementation provides a foundation for further research on room-specific BLE deployment, acoustic coding for noisy classrooms, risk-based device replacement, large-class concurrency, and proximity-aware educational services.

1.7 Methodology Overview

An iterative Agile approach was adopted because the project contained interacting mobile, signal-processing, backend, security, deployment, and interface components. Agile development emphasises working software and response to change (Beck et al., 2001). In this project, each increment was tested on physical Android devices and revised when field behaviour differed from the initial assumption.

Development began with authentication, session management, and a basic mobile workflow. Acoustic transmission and decoding were then implemented through native Android code, followed by lecturer-device BLE advertising and student BLE scanning. Device binding, duplicate prevention, replay records, reports, export, hosted deployment, registered room beacons, signal-source selection, runtime permission guidance, foreground broadcasting, and interface refinement were added in later iterations.

The final verification combined automated and physical tests. The backend API suite exercised authentication, authorisation, proof integrity, multiple-student use of a shared classroom nonce, duplicate prevention, replay scope, compact acoustic proof, beacon-room validation, RSSI rules, device uniqueness, profile privacy, and token revocation. Flutter tests covered authentication rendering, small-screen layouts, and preservation of the installation identifier after logout. Native Android compilation verified the foreground service and Kotlin signal modules. Physical tests used a Redmi 13C as the principal lecturer phone and a POCO C71, Samsung A05, and Vivo Y33 as student phones.

Chapter Three describes the architecture, requirements, proof construction, validation pipeline, database design, and development process in detail.

1.8 Limitations of the Study

The physical evaluation involved four Android phones and repeated small-scale trials rather than a full class. The results therefore demonstrate feasibility on the tested devices but

do not establish performance across all Android manufacturers, operating-system versions, room types, or class sizes.

Acoustic decoding is affected by speaker output, microphone response, phone placement, background noise, and high-frequency attenuation. In the observed tests, useful reception was generally limited to approximately 1-30 cm, with a best observed distance slightly above 50 cm in a quiet environment at high volume. The acoustic path cannot presently cover a medium or large classroom.

BLE provided substantially better coverage, including detections above 10 m, but RSSI remained variable and cannot prove an exact physical distance. Radio signals may also cross a wall or reach an adjacent room. The system mitigates this limitation with room assignment, session status, beacon registration, source freshness, and an RSSI threshold; these controls reduce ambiguity but cannot eliminate relay attacks or every adjacent-room condition.

The installed DX-CP27 beacon was detectable, but lecturer-device BLE was selected more often during the reported tests. Formal tests of multiple beacons, adjacent active rooms, beacon placement, crowd absorption, and simultaneous sessions remain limited. Device binding discourages use of one phone for several student accounts, but it cannot prove that the registered owner is the person carrying the phone.

The backend runs on a free hosted service during development and may require approximately one to two minutes to wake after inactivity. A free service is suitable for development and demonstration but does not provide the availability, monitoring, backup, and support expected of an institution-wide production deployment.

1.9 Organization of the Study

The report is organised into five chapters. Chapter One presents the background, problem, aim, objectives, research questions, scope, significance, methodology overview, and

limitations. Chapter Two reviews manual and automated attendance methods, acoustic communication, BLE proximity, related attendance systems, and fraud controls. Chapter Three explains the development method, requirements, architecture, proof formats, backend validation, database, user workflows, and implementation tools. Chapter Four presents implementation evidence, automated verification, physical test conditions, results, and discussion. Chapter Five summarises the work, states the conclusions, identifies contributions and limitations, and presents recommendations for deployment and further research.

CHAPTER TWO

LITERATURE REVIEW

2.1 Attendance Management in Higher Education

Attendance management covers the capture, validation, storage, retrieval, and reporting of participation in a scheduled activity. Within a university, the record may be used for course administration, student support, accreditation evidence, or compliance with departmental policy. The value of the record depends not only on whether a name appears on a list, but also on whether the record can be attributed to the correct student, class, place, and time.

Traditional attendance methods include roll call, signatures, and paper registers. Their main advantage is low technical complexity. They can be used without a network, smartphone, or specialist equipment. Their limitations become more visible as class size increases: roll call interrupts instruction, a register must circulate through the room, paper records require later compilation, and an absent student may be represented by a colleague. The administrative problem is therefore both operational and evidential. A faster process is useful only if the resulting record remains trustworthy.

Smart attendance systems introduce electronic identification, sensing, mobile applications, or networked databases into this process. Rashid (2024) identifies RFID, NFC, BLE, and related technologies as common foundations for smart-campus attendance. Automation can reduce transcription, support timestamps, and simplify report generation. It can also create new weaknesses. A system that accepts a static identifier without checking its context may digitise proxy attendance rather than prevent it.

2.2 Mobile Attendance Systems

A mobile attendance system uses a smartphone as the principal user terminal. Students may scan a visual code, detect a wireless beacon, provide a biometric sample, or submit location evidence. Lecturers may create sessions, release a temporary credential, monitor responses, and retrieve reports. A remote server can perform authentication and preserve a consistent record across devices.

Smartphones reduce the requirement for a dedicated reader because they already contain wireless radios, cameras, microphones, speakers, clocks, and network interfaces. This advantage is important in settings where the cost of installing and maintaining fixed equipment in every room would be difficult to justify. Lodha et al. (2015) used Bluetooth Smart to reduce the time and human error associated with manual attendance, illustrating the operational value of a wireless attendance workflow.

Mobile deployment also introduces device diversity. Android phones differ in audio frequency response, Bluetooth chipset, antenna arrangement, operating-system version, battery restrictions, and permission behaviour. A design that works on one phone may not produce the same signal level or scan result on another. Mobile software must therefore detect unavailable hardware and disabled services, request permissions at the point of need, present understandable recovery guidance, and avoid deciding attendance exclusively on the client.

The distinction between capture and validation is central to the present study. The phone captures acoustic or BLE evidence, but the server determines whether that evidence is acceptable for the authenticated student and open session. A modified client should not be able to declare itself present merely by displaying a successful scan message.

2.3 Identity, Presence, and Proximity

Identity verification answers the question, "Which account is making the request?" Proximity verification asks whether the device observed evidence associated with the classroom. The two questions are related but not interchangeable. Correct credentials do not establish that their owner is present, and a detectable signal does not establish who is holding the receiving phone.

Time also forms part of attendance evidence. A static classroom identifier can be recorded and reused unless the server checks whether a relevant class is open. A rotating signal reduces this risk by including a short-lived timestamp or random value. Duplicate rules prevent a student from creating several records for one session, while replay rules prevent the same captured evidence from being reused improperly.

Nwabuwe et al. (2023) addressed this wider problem by combining dynamic QR codes, geofencing, and IMEI checking. Their work is relevant because it treats attendance fraud as a combination of code sharing, location, and device identity rather than a single-interface problem. The use of IMEI in that study is not directly transferable to a modern third-party Android application, where hardware identifiers are restricted. The present project instead uses an application installation identifier registered to the student account. This identifier is less authoritative than hardware attestation, but it can prevent casual switching between accounts on the same installation.

2.4 Existing Attendance Technologies

2.4.1 RFID and NFC

RFID uses a reader to detect a radio-frequency tag. NFC is a related short-range technology commonly available in cards and some smartphones. These approaches can make attendance faster than roll call because a tag or card is presented electronically. Miao et al.

(2020) went beyond simple tag detection by analysing passive RFID phase characteristics to distinguish targets and address attendance cheating.

RFID and NFC still require an issued credential, a compatible phone, or a reader at the attendance point. A student may forget or exchange a card, equipment must be installed and maintained, and a congested reader may slow a large class. Their suitability depends on whether the institution already operates an appropriate card infrastructure.

2.4.2 QR Code and Geofencing

QR codes are inexpensive to generate and easy to scan. Ayop et al. (2018) combined QR codes with GPS and event information to create a location-aware attendance process. A dynamic QR code can expire quickly and is safer than a permanent printed code. It can nevertheless be photographed or transmitted electronically, while GPS accuracy may not separate adjacent indoor rooms.

Nwabuwe et al. (2023) reduced these weaknesses by encoding venue boundaries, changing the QR code, and checking the device identity. The study demonstrates that a visual token becomes more credible when it is combined with context. It also shows the implementation cost of relying on several separate checks and the continuing need to handle legitimate device changes.

2.4.3 Biometrics

Biometric attendance associates a record with a physical trait such as a fingerprint or face. Its strength is identity verification rather than proximity signalling. In a controlled installation, biometrics can make credential sharing more difficult. The trade-offs include collection of sensitive personal data, consent and retention policy, camera or sensor quality, environmental lighting, throughput, false rejection, accessibility, and the cost of matching infrastructure.

Face capture was explored during an earlier development iteration of this project but was removed from the active design. Reliable facial comparison would require a validated recognition model, liveness detection, secure biometric-template management, and testing across lighting, camera, pose, and demographic conditions. A basic mobile face detector identifies a face-like region; it does not by itself establish identity. Retaining that experimental feature would have overstated the security of the system and increased privacy risk.

2.4.4 GPS and Network Presence

GPS is effective for many outdoor location services, but satellite reception and accuracy are limited indoors. A geofence around a campus building may show that a device is near the venue without proving that it is in a particular classroom. Wi-Fi association can similarly show that a device is connected to a network whose signal reaches the user. Neither observation necessarily establishes exact room presence, especially where an access point serves several rooms.

For these reasons, the implemented project does not accept GPS or Wi-Fi as an attendance proof. They remain useful comparison technologies in the literature, but the active design is limited to acoustic and BLE evidence that is tied to an open session.

2.4.5 Bluetooth Attendance

Bluetooth attendance has been studied in both tag-based and smartphone-based forms. Lodha et al. (2015) used Bluetooth Smart electronic tags, while Noguchi et al. (2015) developed a student attendance system using BLE beacons and Android devices. Azmi et al. (2018) proposed the UNITEN Smart Attendance System using BLE beacons and compared it with paper and RFID processes. These studies support the feasibility of BLE for attendance and show that beacons can reduce manual interaction.

Bluetooth avoids the optical line-of-sight requirement of QR scanning and is less affected by audible classroom noise than an acoustic signal. It does not provide a precise room boundary. BLE transmissions may travel through doors or walls, and an identifier may be copied. The technology therefore supports proximity evidence, not automatic proof of presence without additional controls.

2.5 Acoustic Communication for Copresence

Acoustic data communication encodes information into a waveform played by a speaker and captured by a microphone. Near-ultrasonic systems operate close to the upper limit of human hearing, typically above the dominant range of speech and music. Their attraction for copresence applications is that ordinary phones already provide the transmitter and receiver.

Getreuer et al. (2018) implemented a near-ultrasonic protocol between consumer devices in the 18.5-20 kHz region. Their design used direct-sequence spread spectrum and short token exchange, and their experiments demonstrated reliable communication at distances that exceeded those obtained by the prototype in this study. This difference is important. The published protocol used a more sophisticated modulation and synchronisation method than the dual-frequency frame used in the project. Its performance cannot be transferred directly to the present implementation.

Jia et al. (2022) combined near-ultrasonic chirps and signal-processing methods for smartphone-based social-distance measurement. The study reinforces the feasibility of acoustic ranging and also illustrates the effort required to handle multipath, non-line-of-sight conditions, channel access, and device timing. An attendance token decoder does not need the same ranging accuracy, but it faces related propagation and hardware constraints.

Near-ultrasonic signalling is not guaranteed to be inaudible on every device. A phone speaker may generate lower-frequency artefacts, and younger users may hear frequencies that

others do not. Increasing amplitude may improve reception but can also increase audible distortion and user discomfort. The signal must consequently be tuned within safe device output and tested with the intended hardware.

The acoustic frame in the present system contains a compact session identifier, generation time, and random challenge. It is encoded into a sequence of tones with start and stop guards. On the receiving phone, the microphone samples are filtered and evaluated in windows to distinguish the expected tone pair. A checksum or digest at proof level prevents a changed payload from being accepted as authentic evidence. Physical tests show that the implementation works, but only at very short range under the conditions recorded in Chapter Four.

2.6 Bluetooth Low Energy Advertising

BLE is designed for low-energy wireless operation in the 2.4 GHz industrial, scientific, and medical band. An advertiser periodically transmits packets on designated advertising channels, and a scanner listens for nearby advertisements. Attendance discovery does not require a conventional paired connection. This reduces interaction time and makes it possible for several student phones to observe the same lecturer broadcast.

The Android permission model affects practical BLE operation. Applications targeting Android 12 or later require runtime permission for scanning, advertising, or connecting, depending on the operation (Android Developers, n.d.). Older Android versions may also associate BLE scanning with location permission. A missing permission can therefore resemble a radio-range problem. This was observed during project development: BLE reception improved when the required permission was enabled.

The lecturer-device path in the present system advertises a session-specific nonce. The nonce rotates during an open attendance period and carries a generation timestamp. A

student submission is accepted only if the nonce belongs to the selected session, remains within the configured freshness window, and has not been improperly reused for that student.

The fixed-beacon path works differently. Commercial room beacons typically broadcast a configured iBeacon or Eddystone identifier repeatedly. They cannot generate a new server nonce for every lecture unless they are reconfigured or connected to an additional controller. The static identifier therefore proves only that a registered beacon was observed. The server supplies the missing context by checking room assignment, current session status, and RSSI policy.

2.7 RSSI and Room Beacons

RSSI is a receiver-side estimate of the power of a detected radio signal. A less negative value generally indicates a stronger observation; for example, -50 dBm is stronger than -80 dBm. The relationship between RSSI and distance is not stable enough to infer an exact classroom position without calibration. Ramirez et al. (2021) showed that orientation, three-dimensional arrangement, and environmental complexity affected BLE measurement and positioning accuracy.

Puckdeevongs et al. (2020) used several BLE stations and fingerprinting methods to estimate student positions in real classrooms. Their system demonstrates that useful indoor positioning is possible, but it required fixed stations, measurements, and computational models. The present project adopts a simpler proximity policy: the app records the strongest acceptable evidence, and the backend rejects a registered beacon below its configured RSSI threshold. This reduces obvious weak-signal acceptance without claiming coordinates.

More than one BLE source may be visible. If the room beacon is substantially stronger than the lecturer phone, selecting the beacon better reflects the local observation. If the lecturer signal is stronger, its short-lived nonce offers better freshness. The app therefore compares valid candidates: it selects one source when that source is at least 10 dB stronger

and retains both sources when their RSSI values are within the 10 dB margin. This rule is an engineering heuristic derived from project testing, not a universal conversion from RSSI to distance.

Room assignment is equally important. A beacon is stored with its UUID or service identifier, major/minor or namespace/instance values, room, active state, and minimum RSSI. When a student submits beacon evidence, the server resolves the detected identity to one active beacon and one open session for that room. It rejects an unregistered beacon, a room mismatch, a weak observation, or an ambiguous room state.

2.8 Attendance Fraud and Security Controls

BLE attendance is vulnerable when a stable beacon message is accepted without additional evidence. Kim et al. (2018) analysed a deployed BLE beacon attendance system and demonstrated signal imitation and forwarding attacks. Their findings show that physical beacons should not be treated as secrets: an advertisement is intentionally observable by nearby devices and may be copied.

The present design uses several controls in response:

1. lecturer acoustic and BLE payloads are short-lived and tied to a session;
2. proof data includes a SHA-256 digest that the backend recomputes;
3. only an authenticated student can submit for the student identity associated with the token;
4. each student may create at most one proof for a session;
5. replay records are scoped to the session and student, allowing classmates to use the same broadcast while preventing improper reuse by one student;
6. a room beacon must be registered, active, assigned to the session room, and observed at an acceptable RSSI;
7. only one open session is permitted for a room at a time; and

8. a student account is associated with one installation identifier unless an administrator performs a controlled reset.

These controls address casual sharing, duplicate submission, stale capture, cross-room beacon use, and simple client tampering. They do not prevent every adversarial scenario. A student could physically carry a friend's registered phone, an advanced attacker could relay a live signal in real time, and an application-generated installation identifier is not hardware-backed proof. Security claims must therefore remain proportional to the implementation and evaluation.

2.9 Review of Closely Related Studies

2.9.1 Bluetooth Smart Attendance

Lodha et al. (2015) proposed a Bluetooth Smart attendance system using electronic tags and a Bluetooth-enabled device. The system reduced the time and human errors associated with manual attendance and produced records for administrative use. Its reliance on tags differs from the present student-phone scanning model, and the published description does not address a rotating session credential or room-specific replay controls.

2.9.2 Android and BLE Beacon Attendance

Noguchi et al. (2015) designed an attendance system using BLE beacons and Android devices. The work is closely related because an Android phone detects a classroom beacon before submitting attendance information. A fixed beacon, however, remains a reusable signal unless the server adds session context. The present project extends this pattern with an open-session requirement, room registration, RSSI policy, a lecturer-generated rotating BLE option, and replay checks.

2.9.3 UNITEN Smart Attendance System

Azmi et al. (2018) implemented UniSas with BLE beacons and compared paper, RFID, and beacon attendance using accuracy, time, and energy considerations. The authors reported that their beacon method performed better than the traditional alternatives evaluated. The system provides strong evidence that BLE beacons are practical in university attendance. The present study differs by allowing the lecturer phone and a fixed beacon to act as alternative BLE sources and by documenting the security limitation of a static beacon.

2.9.4 BLE Classroom Positioning

Puckdeevongs et al. (2020) combined BLE stations, RSSI fingerprinting, and neural-network methods to record attendance and estimate classroom position. Their real-classroom evaluation showed that BLE can support attendance despite indoor interference, but it also recorded positioning error and required several installed stations. The study supports the use of BLE in a classroom while confirming that robust position estimation is more complex than reading one RSSI value.

2.9.5 Proximity-Based IoT Attendance

Hayati and Nugraha (2023) implemented an Android smart attendance application that detected a beacon and communicated with a cloud database. Their field evaluation covered two courses and multiple mobile devices; the authors reported functional-test results of 91.67% and 100%, with one limitation arising from a phone that did not detect the beacon. The architecture is close to the present project and is particularly relevant to Telecommunication Science. It also confirms that handset compatibility can affect beacon systems.

2.9.6 Fraud-Aware Mobile Attendance

Nwabuwe et al. (2023) combined dynamic QR codes, geofencing, and IMEI verification. Their evaluation reported successful mitigation of the tested attendance-fraud attempts. Although the signal technologies differ, the study supports dynamic session evidence and one-device controls. The present project substitutes acoustic and BLE proximity for the QR/geofence path and uses an application installation identifier because direct IMEI access is restricted.

2.9.7 BLE Signal Imitation

Kim et al. (2018) examined a commercial BLE electronic attendance system and found that predictable, static beacon messages could be imitated and forwarded. This is the most important security-related prior work for the beacon path. The result explains why the CP27 identifier is never treated as a session secret and why server-side room, time, account, and duplicate rules are required.

2.9.8 Acoustic Copresence

Getreuer et al. (2018) developed a near-ultrasonic token protocol on consumer hardware, while Jia et al. (2022) developed smartphone near-ultrasonic ranging. Neither paper presents a classroom attendance application, but both establish that phones can exchange or measure high-frequency acoustic signals. Their methods also show that modulation, synchronisation, coding, multipath handling, and hardware response determine range. The present project's simpler acoustic implementation is therefore evaluated independently rather than assumed to reproduce their performance.

2.10 Comparative Analysis

Table 2.1
Comparison of Closely Related Studies

Study	Principal technology	Reported contribution or result	Limitation relevant to this project
Lodha et al. (2015)	Bluetooth Smart tags	Reduced manual attendance time and human error	Requires electronic tags; limited discussion of rotating proof and fraud controls
Noguchi et al. (2015)	BLE beacon and Android	Classroom beacon detection and mobile attendance submission	Static beacon context can be reused unless constrained by session rules
Azmi et al. (2018)	BLE beacon	UniSas implementation; reported better evaluated performance than paper and RFID alternatives	Beacon-centric design; static-signal security not the main focus
Kim et al. (2018)	BLE security analysis	Demonstrated imitation and forwarding against a deployed attendance system	Analyses attack rather than implementing this project's combined workflow
Puckdeevongs et al. (2020)	Multiple BLE stations and RSSI fingerprinting	Real-classroom attendance and indoor-positioning evaluation	Requires fixed infrastructure, calibration, and positioning models
Hayati and Nugraha (2023)	Android, beacon, cloud database	Field-tested proximity attendance across two courses; 91.67% and 100% functional results reported	One device failed to perceive the beacon; no acoustic alternative
Nwabuwe et al. (2023)	Dynamic QR, geofence, IMEI	Combined dynamic proof, location, and device checks to mitigate tested fraud cases	GPS/geofence does not reliably separate indoor rooms; IMEI access is restricted
Getreuer et al. (2018)	Near-ultrasonic communication	Demonstrated robust short-token communication on consumer hardware	Not an attendance system; uses a more advanced protocol than this prototype
Jia et al. (2022)	Near-ultrasonic ranging	Demonstrated smartphone acoustic proximity measurement	Ranging algorithm and test objective differ from attendance-token decoding
Present study	Acoustic token, lecturer BLE, registered room beacon, Django validation	Combines alternative physical-signal paths, room-aware beacon validation, device binding, replay control, reporting, and hosted access	Small four-device evaluation; acoustic range is very short; multi-room scale remains unproven

2.11 Research Gap

The literature establishes several individual capabilities: Bluetooth can reduce attendance time; Android phones can detect classroom beacons; BLE stations can support indoor positioning; near-ultrasonic signals can exchange short tokens; and dynamic

credentials, location, or device checks can reduce selected forms of fraud. It also identifies a serious weakness: a static BLE attendance signal can be copied or forwarded.

Within the reviewed work, the remaining practical gap is not the absence of another attendance interface. It is the need for a modest, deployable system that combines:

1. a rotating lecturer-device signal for freshness;
2. a fixed room beacon for coverage where required;
3. an independent acoustic path for close-range copresence;
4. room-aware server validation rather than unconditional beacon acceptance;
5. proof integrity, duplicate, replay, and device-association controls; and
6. complete lecturer and student workflows with session-specific reporting.

The implemented study addresses this gap through alternative acoustic and BLE proof paths rather than requiring both signals for every submission. This choice reflects observed reliability: BLE is the main classroom mechanism, while acoustic evidence is retained for very short-range use. The contribution is the integration and validation policy, not a claim that the underlying radio or acoustic technologies are new.

2.12 Summary

The reviewed literature shows that electronic attendance can improve speed and record management, but each technology has a different evidential limit. RFID and NFC depend on credentials and readers; QR codes can be shared; GPS and network presence are weak at room level; biometrics introduce privacy and operational concerns; BLE offers useful classroom reach but can be variable, copied, or relayed; and acoustic communication depends strongly on coding, hardware, distance, and noise.

The literature supports a design in which BLE provides practical proximity coverage, acoustic signalling provides supplementary copresence evidence, and the backend evaluates every proof within an authenticated student, open session, room, time, and device context.

Chapter Three explains how those principles were translated into the implemented architecture.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 Development Methodology

The system was developed with an iterative Agile approach. Agile was selected because the project combined a mobile interface, native Android services, acoustic signal processing, BLE communication, server-side validation, a relational database, deployment, and physical-device testing. These parts could not be evaluated adequately through a single sequential implementation. The Agile emphasis on working software and response to change allowed each increment to be tested before the next design decision was fixed (Beck et al., 2001).

The development process consisted of repeated cycles:

1. define the behaviour required for one workflow;
2. implement the mobile and backend components;
3. run focused automated checks;
4. test the feature on physical Android devices;
5. record the failure or usability problem;
6. revise the design; and
7. retain the verified implementation in version control.

This process materially changed the system. Acoustic signalling was initially expected to provide wider coverage, but device testing showed that the implemented decoder was dependable only at close range. BLE reception improved after the required Android permissions were handled correctly and consequently became the principal classroom mechanism. The DX-CP27 beacon was introduced as a room-specific BLE source, after which room assignment, beacon parsing, RSSI selection, and adjacent-room conflict rules were added.

The final implementation increment added a native Android foreground service, persistent notification, and partial wake lock so that a lecturer broadcast continues when the lecturer changes pages or locks the screen. This revision arose directly from physical use rather than from the original interface specification.

3.2 Analysis of the Existing Attendance Process

The existing manual process can be represented as follows:

1. the lecturer announces attendance;
2. students respond to roll call or sign a circulated register;
3. the lecturer or class representative checks the list;
4. the record is stored on paper or transferred to another system; and
5. attendance is later compiled for the course.

This process has four principal weaknesses. First, attendance collection competes with lecture time. Second, the record is vulnerable to proxy signing or response. Third, paper records require manual storage and compilation. Fourth, there is limited evidence linking the entry to a particular student, session, room, and time.

A basic web form improves storage but does not solve the evidence problem. A student outside the room can submit a shared link if the server only checks credentials. A static QR code or beacon identifier presents a similar weakness if it can be photographed, recorded, copied, or relayed. The proposed system therefore treats proximity capture and backend validation as separate stages.

3.3 Proposed System

The proposed system is a hosted client-server application with three operating roles: lecturer, student, and administrator. The lecturer and student use one Flutter Android

application, while the role returned by the backend determines the portal shown after login. The administrator uses the Django administration interface.

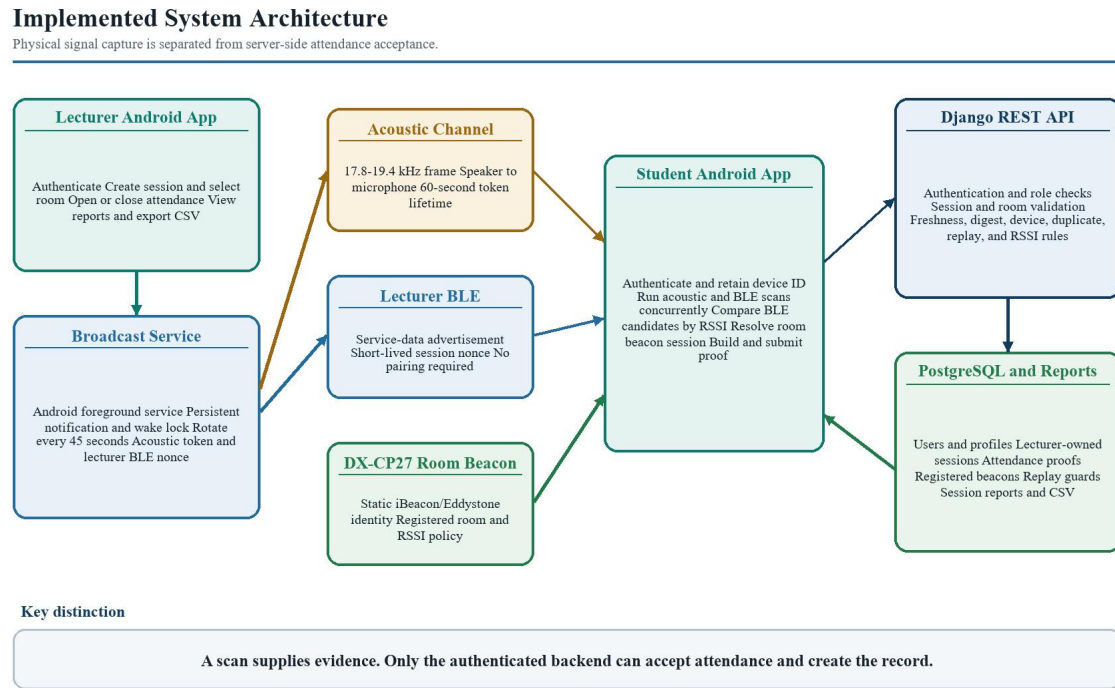
The lecturer creates a session containing the course code, course title, room, and scheduled times. The backend supplies the lecturer name from the authenticated profile rather than trusting a typed name. When attendance is opened, the backend sets the opening time and a closing time 15 minutes later. Only one attendance session may be open for a room at a time.

The lecturer phone starts a foreground service that broadcasts acoustic and BLE evidence. The service generates new values every 45 seconds. Each generated acoustic token and BLE nonce remains acceptable for up to 60 seconds, giving a 15-second overlap for a scan or request in progress. The service continues across navigation and screen lock, but it stops when attendance is closed, the service is explicitly stopped, or the application task is removed.

The student initiates one scan operation. Acoustic capture and BLE scanning run concurrently to reduce waiting time. The app parses evidence from the lecturer phone and any supported room beacon. It compares valid BLE candidates by RSSI, resolves a fixed beacon to the currently open room session through the API, presents a concise result, and enables submission only when at least one acceptable path is available.

The student submits a proof containing the session, student identity, installation identifier, captured evidence, observed RSSI, observation time, and SHA-256 digest. The backend repeats all material checks and creates the attendance record within a database transaction.

Figure 3.1
Implemented System Architecture



Note. Developed by the researcher from the implemented mobile, native Android, API, and database components.

3.4 System Requirements

3.4.1 Functional Requirements

The functional requirements are grouped by role.

Lecturer requirements

1. register and authenticate as a lecturer;
2. create, update, select, and permanently delete an owned session;
3. select a room from registered beacon-room information;
4. open or close attendance for an owned session;
5. start and stop acoustic and lecturer-device BLE broadcasting;
6. retain broadcast operation while navigating within the app or locking the screen;
7. search sessions and attendance records;
8. view a session-specific attendance report; and

9. export the report as CSV.

Student requirements

1. register and authenticate with a unique matric number;
2. maintain one installation identifier across logout and subsequent login;
3. receive microphone, Bluetooth, nearby-device, and location guidance where required;
4. scan acoustic and BLE channels concurrently;
5. identify whether BLE evidence came from the lecturer phone, a room beacon, or both;
6. resolve a registered beacon to an open session in its assigned room;
7. review a clear successful or unsuccessful scan state;
8. submit no more than one attendance proof for a session; and
9. view attendance history and profile/device information.

Administrator requirements

1. manage users and profiles;
2. inspect sessions, proofs, replay records, and registered beacons;
3. configure beacon identity, room, RSSI threshold, transmit power, and interval;
4. reset a student's device binding through a controlled admin action; and
5. preserve submitted attendance and replay records as read-only audit data.

3.4.2 Non-Functional Requirements

Security. API operations require token authentication except registration, login, and health checking. Role and object ownership are enforced on the server. Production settings support HTTPS redirection, secure cookies, HTTP Strict Transport Security, and environment-managed secrets.

Integrity. The backend recomputes the proof digest and stores proof and replay records atomically. Database uniqueness constraints provide a final guard against duplicate attendance, shared device bindings, and repeated use of a rotating signal by one student.

Usability. The mobile interface uses role-specific navigation, concise status text, focused forms, searchable lists, expandable student rows, and recovery messages that do not expose exception traces.

Availability. The hosted API removes dependence on a changing local IP address. The free hosting tier may introduce a cold start and is treated as a development deployment rather than an institutional service-level guarantee.

Compatibility. The completed signal implementation targets Android. Layout tests use a 390 x 844 logical-pixel viewport to detect mobile overflow.

Maintainability. Mobile networking is centralised in a shared API client, proof formats are isolated in a codec, and server validation is concentrated in serializers and role-aware views. Database changes are versioned through Django migrations.

Privacy. The active API does not collect biometric evidence. Profile responses contain only the identity, role, and registered-device information required by the attendance workflow.

3.5 System Architecture

The architecture in Figure 3.1 contains five logical layers.

3.5.1 Presentation Layer

Flutter provides the authentication screen and the lecturer and student portals. The student portal contains Scan, History, and Profile destinations. The lecturer portal contains Session, Live, Reports, and Profile destinations. Navigation preserves the active student scan state, and lecturer broadcasting is not owned by a page widget.

3.5.2 Native Android Signal Layer

Flutter communicates with Kotlin code through a method channel. The native layer contains:

1. AttendanceBroadcastService for foreground operation and 45-second rotation;
2. AcousticFrameEncoder and AcousticTransmitter for PCM generation and playback;
3. AcousticFrameDecoder for microphone capture, filtering, guard detection, tone decisions, frame parsing, and diagnostics; and
4. BleAdvertiser for lecturer-device service-data advertising.

BLE scanning is performed through the Flutter BLE package because scan results must be integrated with Flutter state and the room-beacon parser.

3.5.3 Physical Signal Layer

Three physical observations are supported:

1. an acoustic frame emitted by the lecturer speaker;
2. a short-lived BLE service-data advertisement emitted by the lecturer phone; and
3. an iBeacon or Eddystone UID frame emitted by a fixed room beacon.

The first two carry session-specific rotating data. The room beacon carries a configured identity and depends on backend room/session validation.

3.5.4 API and Validation Layer

Django REST Framework exposes authentication, session, proof, room-beacon, report, export, and health endpoints. Token authentication identifies the account. Serializers parse and validate proof data, while views apply role, ownership, device-binding, room, and query-scope rules.

3.5.5 Data and Reporting Layer

The relational database stores users, profiles, sessions, proofs, registered beacons, and replay records. Lecturer queries are restricted to sessions created by that lecturer. Student history is restricted to the authenticated identity. Reports are filtered by the selected session,

and CSV export contains serial number, student name, matric number, device identifier, and signal mode.

3.6 User Roles and Authorisation

3.6.1 Lecturer

A lecturer may create and manage only sessions associated with the lecturer profile. The name stored on a session is taken from the authenticated account. Another lecturer cannot update, delete, open, close, report on, or export that session. These controls are applied in the API and are not dependent on hidden interface buttons.

3.6.2 Student

A student may view only currently open sessions and the student's own proof history. The `student_id` in a proof must equal the matric number or username of the authenticated profile. The server does not permit a student to submit an arbitrary matric number.

3.6.3 Administrator

The administrator may configure operational records through Django admin. Proofs and replay records are read-only because modifying or deleting accepted evidence would weaken auditability. A device binding may be reset through a named administrative action when a legitimate replacement is approved.

3.7 Session Lifecycle

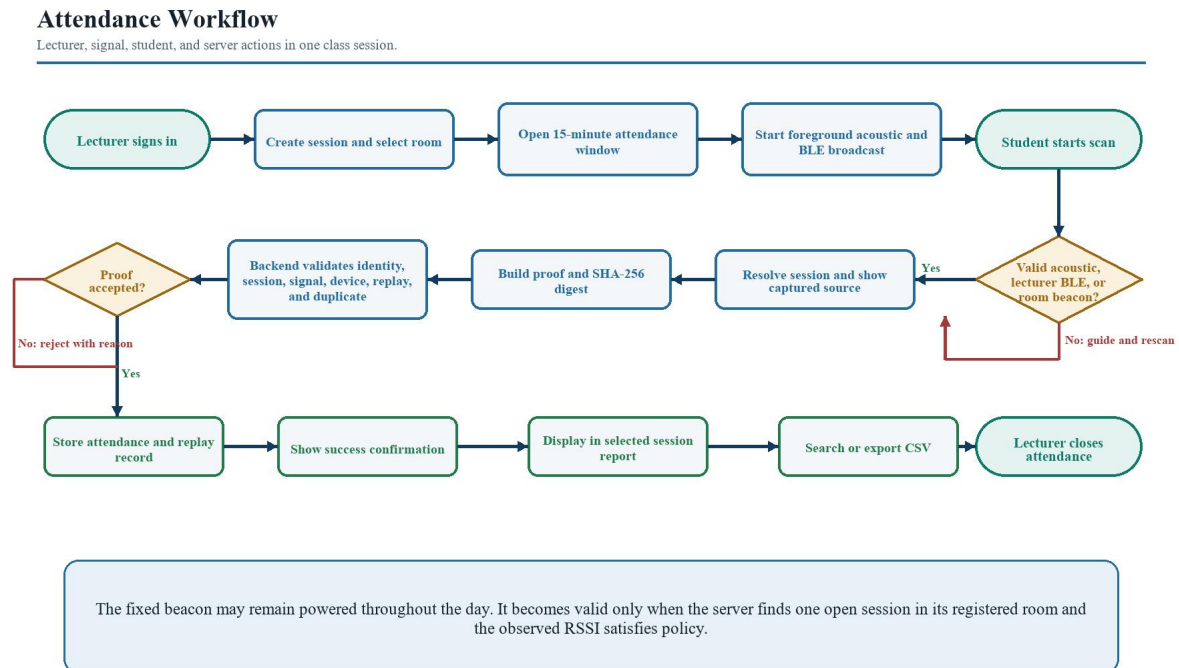
A session moves through the following states:

1. **Created:** the lecturer supplies course and room information; attendance is closed.
2. **Open:** the owner opens attendance; the backend records `attendance_opened_at` and computes `attendance_closes_at`.

3. **Broadcasting:** the Android foreground service transmits rotating evidence for that session.
4. **Closed:** the lecturer closes attendance or the configured 15-minute window expires.
5. **Inactive or deleted:** the lecturer may deactivate or permanently delete the owned session.

Before returning session or room data, the backend closes any record whose attendance closing time has passed. A room-conflict check prevents another session from opening in the same named room. A fixed beacon remains physically active, but it cannot resolve to a session before state 2 or after state 4.

Figure 3.2
Attendance Workflow



Note. Developed by the researcher to show lecturer, student, signal, and server actions within one attendance session.

3.8 Acoustic Proof Design

3.8.1 Compact Token

The current acoustic token has the following form:

```
ac2|session_base36|issued_epoch_base36|challenge
```

Base-36 encoding shortens the session and timestamp fields. The challenge is an eight-character random string generated with `SecureRandom`. The token omits descriptive course data because the backend can retrieve that information from the session identifier.

3.8.2 Frame Encoding

The encoder uses binary frequency-shift keying. Its principal parameters are shown in Table 3.1.

Table 3.1
Acoustic Encoder Parameters

Parameter	Implemented value
Sampling rate	44,100 samples/s
Start guard	17,800 Hz for 140 ms
Binary zero	18,400 Hz
Binary one	18,900 Hz
Stop guard	19,400 Hz for 140 ms
Bit duration	12 ms
Preamble	10101010
Length field	8 bits
Integrity byte	XOR checksum
Amplitude coefficient	0.42
Frame repetitions	5
Inter-frame gap	18 ms

The encoder converts the UTF-8 token into bytes, prefixes the preamble and payload length, appends an XOR checksum, and generates a tapered sine wave for each bit. A four-millisecond amplitude ramp reduces abrupt edges at tone boundaries.

3.8.3 Decoding

The decoder records at 44.1 kHz for up to six seconds. A high-pass filter at 17.2 kHz and low-pass filter at 19.7 kHz isolate the intended band. Goertzel-style tone-energy measurements are evaluated in 12 ms windows. Start and stop guards, the preamble, length, payload, and checksum are then checked.

Diagnostics distinguish low ultrasonic energy, missing guards, ambiguous bit decisions, incomplete frames, and checksum failure. These messages support testing but are translated into concise user-facing guidance in the app.

The acoustic token is not accepted on local decode alone. The backend checks the encoded session, issue time, challenge, proof digest, authenticated student, session status, duplicate record, and replay record.

3.9 BLE Proof Design

3.9.1 Lecturer-Device BLE

The lecturer BLE value has the form:

`ble|session_id|issued_epoch|nonce`

It is placed in service data under the project service UUID. The foreground service rotates the nonce at the same 45-second interval as the acoustic challenge. The student parser validates the four fields and evaluates local freshness before enabling submission. The server enforces the 60-second signal lifetime independently.

3.9.2 Registered Room Beacon

Two beacon formats are supported:

`beacon|ibeacon|uuid|major|minor`

`beacon|eddystone_uid|namespace_id|instance_id`

The CP27 can expose several frames, but the project registers only the selected iBeacon or Eddystone UID identity. A `RegisteredBeacon` record stores the readable beacon name, room, format fields, reference RSSI, minimum accepted RSSI, transmit power, advertising interval, active state, and notes.

The fixed beacon does not carry a session nonce. A student first sends the observed beacon identity and RSSI to the resolve endpoint. The server finds a unique active beacon, confirms its room, closes expired sessions, and returns the one open session for that room. The final proof repeats the same identity so validation is not based only on the earlier lookup.

3.9.3 BLE Source Selection

The scanner collects nearby results for ten seconds, deduplicates advertisements, and identifies the strongest valid lecturer payload and strongest recognised beacon payload. Selection follows three cases:

1. if one source is at least 10 dB stronger, select that source;
2. if both are present within the 10 dB margin, retain both sources; or
3. if only one valid source is present, use that source.

The selected RSSI is the stronger observation. When both sources are retained, the final proof contains both the current lecturer nonce and beacon identity. The server validates each supplied source.

3.10 Proof Construction and Integrity

The student proof contains:

1. session identifier;
2. authenticated student identifier;
3. installation device identifier;
4. acoustic token, if captured;

5. lecturer BLE nonce, if captured;
6. room-beacon proof, if captured;
7. RSSI value;
8. beacon RSSI where available;
9. UTC observation time; and
10. SHA-256 digest.

The digest input is the pipe-separated sequence of session, student, device, acoustic token, BLE nonce, a reserved compatibility field, beacon proof, RSSI, and the exact observation-time string sent by the client. The backend reconstructs the same sequence and compares digests with a constant-time function.

This digest detects accidental corruption and simple field editing between proof construction and API validation. It is not a digital signature because no secret key is held in secure hardware on the client. A determined attacker who controls a modified client can recompute SHA-256. The decisive protections remain authenticated identity, session ownership, signal parsing, freshness, device binding, database constraints, and server policy.

3.11 Backend Validation Pipeline

The validation order is designed to reject invalid context before creating any record:

1. confirm that the request timestamp is no more than 120 seconds old and not more than 10 seconds in the future;
2. confirm that the session is active, open, owned by a lecturer, and within its closing time;
3. require at least one supported acoustic, lecturer BLE, or room-beacon source;
4. parse every supplied proof format;
5. compare the acoustic or lecturer BLE session identifier with the selected session;
6. enforce the 60-second lifetime for rotating signals;
7. resolve and validate a supplied fixed beacon, room, and RSSI;

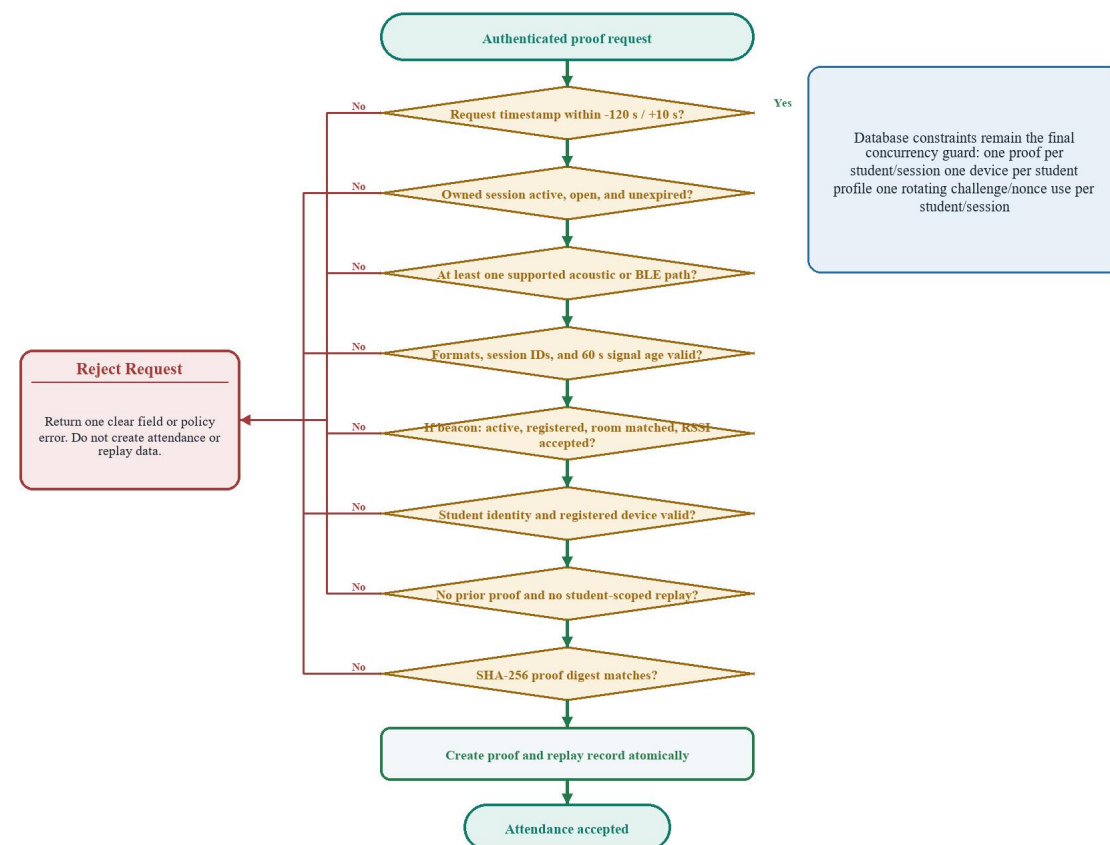
8. compare submitted student identity with the authenticated account;
9. reject prior use of the same rotating challenge or nonce by that student in the session;
10. reject an existing attendance record for that student and session;
11. recompute the SHA-256 digest;
12. verify or bind the installation identifier; and
13. create the proof and replay record atomically.

The final uniqueness constraints handle requests that pass application checks concurrently. An integrity error is translated into an understandable duplicate/replay response.

Figure 3.3
Backend Attendance Validation Flow

Backend Validation Flow

Every material decision is repeated on the server.



Note. Developed by the researcher from the implemented serializer, model, and database validation rules.

3.12 Device Binding and Fraud Controls

On first valid student use, the installation identifier may be bound to the profile. The database permits one non-empty identifier for only one student profile. The identifier is preserved when the user logs out; deleting it during logout would cause the legitimate installation to appear new at the next login. A second student cannot register or bind the same identifier.

Device binding addresses one scenario: a student logging into several accounts on the same application installation. It does not prevent a student from carrying a friend's already registered phone into class. It also does not equal Android hardware attestation. The report therefore describes device binding as a deterrent and trust signal, not as proof of personal identity.

Duplicate attendance is enforced by a unique pair of session and student. Replay constraints are deliberately student-scoped. Every student in a classroom may legitimately receive the same broadcast nonce, so a global uniqueness rule would accept only the first student. The implemented constraints instead prevent the same student from using the same acoustic challenge or BLE nonce twice within one session.

Other controls include lecturer ownership, one open session per room, fixed-beacon registration, minimum RSSI, signal expiry, proof digest, token revocation at logout, password validation, and read-only audit records.

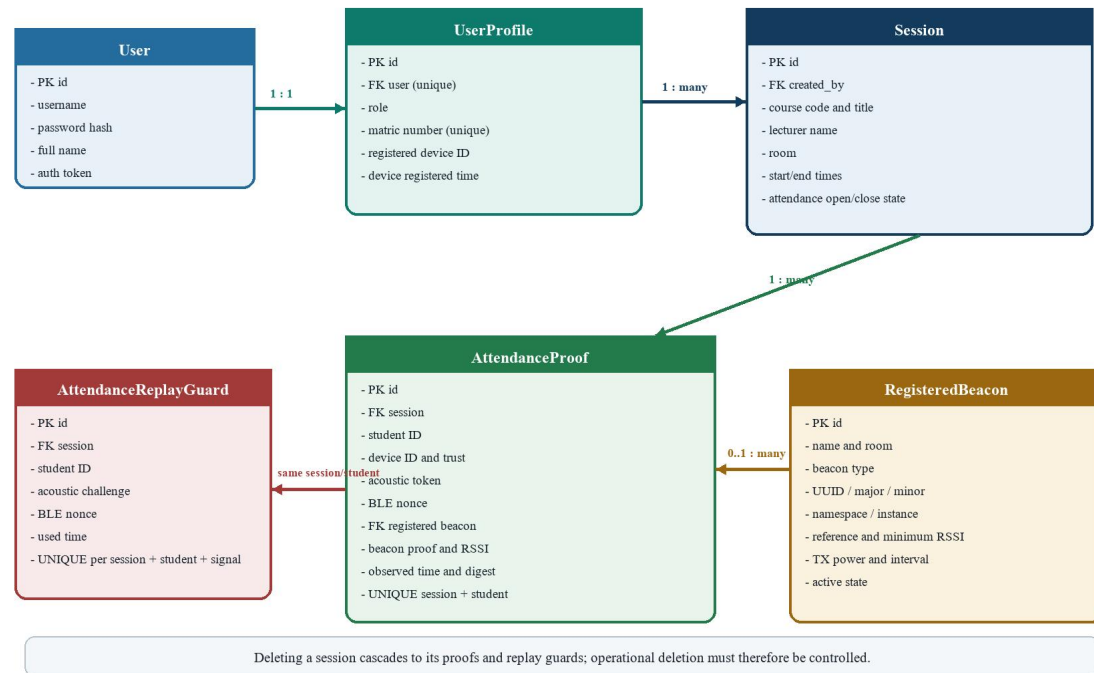
3.13 Database Design

The core data model is shown in Figure 3.4 and summarised in Table 3.2.

Figure 3.4
Core Entity Relationship Model

Core Entity Relationship Model

Active attendance entities, relationships, and database constraints.



Note. Developed by the researcher from the active logical database model.

Table 3.2
Core Database Entities

Entity	Purpose	Selected fields and constraints
User	Django authentication identity	username, password hash, full name, email, active/staff flags
UserProfile	Attendance role and device association	one-to-one user; role; unique matric number; one unique non-empty student device ID
Session	Lecturer-owned class and attendance state	course code/title; room; times; owner; open/close state; 15-minute window
AttendanceProof	Accepted attendance record	one record per session/student; signal fields; device trust; RSSI; observed and creation times
RegisteredBeacon	Fixed room-beacon configuration	iBeacon or Eddystone identity; room; minimum RSSI; transmit settings; active state
AttendanceReplayGuard	Student-scoped rotating-signal use	unique session/student/challenge and session/student/BLE nonce
Token	API authentication	one token associated with a signed-in user; revoked on logout

The logical design presented in Figure 3.4 contains only the entities and fields used by the active attendance workflow. Historical migration compatibility does not alter the accepted acoustic and BLE proof paths.

3.14 Data Flows

3.14.1 Registration and Login

A registrant submits full name, role, identity, password, and installation identifier. Student matric numbers are normalised to uppercase and checked case-insensitively. Django password validators enforce the password policy. User and profile creation occur in a transaction. Login is case-insensitive, returns an API token and profile summary, and loads the role-specific portal.

3.14.2 Session Creation and Opening

The lecturer submits course and room data. The API records the authenticated lecturer as owner. On opening, the API checks room conflict, sets attendance times, and returns the session. The app then starts the foreground service with the first generated acoustic and BLE values.

3.14.3 Student Scan

The student app confirms required permissions and Bluetooth state. Acoustic and BLE scans run concurrently. Local parsing records diagnostic evidence. A fixed beacon is resolved through the backend because its frame does not contain a session. The app retrieves session metadata and enables submission only for a valid current path.

3.14.4 Submission and Reporting

The app constructs and posts the proof. On success, it records the session locally to prevent another immediate submission and displays a success confirmation. The backend

remains the authoritative duplicate guard. The lecturer selects a session to view student rows and may expand a row for matric number, course, room, device, and scan mode before exporting CSV.

3.15 User Interface and Experience Design

The interface uses a restrained mobile visual language rather than decorative dashboards or promotional copy. It applies a consistent colour palette, typography hierarchy, spacing scale, status language, and bottom navigation. Content is centred within a maximum width so that the web build remains usable without turning the Android interface into a desktop-style page.

The lecturer session form requests only information the lecturer must provide. Lecturer name and token version are derived internally. Room selection uses known beacon rooms while allowing the course/session information to remain readable. Active broadcast status is presented in user terms rather than displaying raw nonce values.

The Live and Reports pages use search fields and compact one-line student/session rows. A student row expands on tap to show only the operational details requested for classroom use. This structure scales better than a large card for every student.

The student scan page separates action, evidence, and result. The principal action is a single scan button. Raw acoustic and BLE fields are not intended as manual inputs. A successful scan states the source and session; an unsuccessful scan gives one recovery action, such as enabling Bluetooth, granting permission, moving closer, or scanning again.

Error messages are mapped from network and API exceptions into short descriptions. Stack traces, socket exception text, and internal class names are not shown to users.

3.16 Deployment Design

The Android application uses `https://sa-acoustic-ble.onrender.com` as its default API. The value can still be replaced at build time through `API_BASE_URL` for controlled development, but the installed application does not expose an IP-address editor. This prevents stale LAN settings and removes the need to rebuild whenever a local address changes.

The Django service reads its secret key, debug state, allowed hosts, CSRF origins, database URL, and security settings from environment variables. Development uses SQLite. Hosted operation uses PostgreSQL. Static files are served through the deployment configuration, and a health endpoint allows the hosting platform or tester to confirm that the API is awake.

The shared mobile API client applies a 75-second timeout to tolerate a free-tier cold start and converts timeouts into understandable service-waking guidance. A logout request revokes the server token; local credentials are cleared even if the network is unavailable.

3.17 Software and Hardware Tools

Table 3.3
Development and Test Tools

Category	Tool or component	Use
Mobile framework	Flutter and Dart	Cross-platform UI, state, HTTP services, local storage
Android native layer	Kotlin	Foreground service, audio capture/playback, BLE advertising
Backend	Python, Django, Django REST Framework	Authentication, sessions, validation, reporting, admin
Database	SQLite and PostgreSQL	Local development and hosted relational storage
Hosting	Render	Public Django API deployment
Version control	Git and GitHub	Phase tracking, history, and remote backup
Lecturer test phone	Redmi 13C	Main acoustic and BLE broadcaster
Student test phones	POCO C71, Samsung A05, Vivo Y33	Signal scanning and proof submission

Category	Tool or component	Use
Fixed beacon	DX-CP27 Mini	iBeacon/Eddystone room identity

3.18 Verification Plan

Verification was divided into four levels.

Backend API tests cover authorisation, session ownership, room conflict, session expiry, valid and duplicate BLE proof, shared classroom nonce use by different students, digest tampering, identity mismatch, unsupported proof rejection, compact acoustic proof, beacon room and RSSI validation, ambiguous room state, unique device registration, profile privacy, and logout token revocation.

Flutter tests cover signed-out rendering, student and lecturer layouts at the test mobile viewport, and persistence of the installation identifier after logout.

Native build verification compiles the Android/Kotlin modules, foreground service, BLE advertiser, and acoustic components against the configured Android toolchain.

Physical tests cover registration, login, permission prompts, session creation, foreground broadcast, acoustic detection, BLE distance bands, obstacles, CP27 detection, proof submission, duplicate behaviour, hosted API wake-up, report display, and CSV export. Chapter Four reports only observations actually recorded during these tests.

3.19 Ethical, Privacy, and Security Considerations

The application stores identity, attendance, device association, session, and signal metadata. These records should be collected for attendance administration only and protected by role-based access. An institutional deployment should define retention periods, backup policy, device-change approval, breach response, and a process by which a student can query or correct an attendance record.

The active design avoids biometric collection. The installation identifier is shown in a shortened display form where a full identifier is unnecessary. The API uses HTTPS in hosted operation, passwords are hashed by Django, and secrets are supplied through the environment rather than committed to the repository.

No proximity method should be represented as infallible. BLE can cross room boundaries or be relayed, acoustic decoding can reject a present student, and a registered phone can be carried by another person. These limitations should be disclosed to users and considered in any disciplinary use of the records. Attendance evidence should support academic administration, not replace fair review of legitimate technical exceptions.

3.20 Summary

This chapter described the iterative method and the implemented architecture of the smart attendance system. The system separates signal capture from server acceptance, uses rotating lecturer acoustic and BLE evidence, supports a fixed beacon only within a registered room and open session, and applies identity, digest, freshness, duplicate, replay, device, ownership, and RSSI controls.

The design also addresses operational requirements that became visible during testing: concurrent scanning reduces wait time, a foreground service preserves lecturer broadcast across navigation and screen lock, compact reports support larger lists, and a hosted API removes dependence on a local address. Chapter Four evaluates the resulting implementation through automated checks and four-device physical observations.

CHAPTER FOUR

IMPLEMENTATION, TESTING, RESULTS AND DISCUSSION

4.1 System Implementation

The completed prototype consists of a Flutter Android application, native Kotlin signal modules, a Django REST API, and a relational database. The API is deployed at <https://sa-acoustic-ble.onrender.com>, allowing installed applications to authenticate and submit records without using a local computer address.

The Flutter application presents separate lecturer and student portals after authentication. The lecturer can create a course session, select a room, open attendance, maintain a live broadcast, review session-specific records, search, and export CSV. The student can scan, review the captured mode, submit once, view history, and inspect the registered installation information.

The native Android layer provides the operations that could not be implemented reliably as browser-only functions. Kotlin classes perform acoustic PCM generation, microphone decoding, BLE advertising, and foreground-service management. The lecturer broadcast service maintains a persistent notification, acquires a partial wake lock, and rotates the acoustic and BLE values every 45 seconds. The service is separate from the selected Flutter page, so navigating from Session to Live, Reports, or Profile does not intentionally stop transmission.

The backend performs the authoritative attendance decision. It applies authentication, role, session-owner, room, time, signal, device, digest, duplicate, and replay rules before storing a proof. The Django administration interface exposes operational records while keeping accepted proof and replay data read-only.

4.2 Implemented User Workflows

4.2.1 Lecturer Workflow

The lecturer signs in, creates or selects a session, chooses the classroom, and opens attendance. The API prevents another session from opening in the same room and starts a 15-minute attendance window. The app then requests the required Bluetooth state and permissions before starting the Android foreground service.

The session page presents the course, room, and broadcast state without requiring the lecturer to manage token values. The Live page provides searchable sessions and compact rows, while the Reports page displays students for the selected session and exports the same data in CSV form.

4.2.2 Student Workflow

The student signs in with a matric-number account. The installation identifier is loaded from local storage and preserved across logout. A scan starts acoustic capture and BLE discovery concurrently. When lecturer BLE or room-beacon evidence is found, the app records the source, RSSI, and session. An acoustic decode supplies the compact acoustic token directly.

Submission is disabled unless the scan contains a valid current path. After server acceptance, a confirmation is displayed and the session is added to the local submitted set. This local check improves feedback, while the server and database remain the final duplicate controls.

4.2.3 Reporting Workflow

The lecturer selects a session before retrieving its report. Student entries are shown as one-line rows to avoid allocating a large card to every member of a class. Tapping a row

reveals the matric number, course, room, device, and captured signal mode. Search applies to session, course, room, student name, matric number, and device values.

4.3 Evaluation Design

Evaluation was divided into automated software verification and small-scale physical-device testing. The two sets of results answer different questions. Automated tests verify deterministic rules in the software. Physical tests show whether signal capture and user workflows operated on the available phones under the observed conditions.

The physical tests were repeated over more than one week with one principal lecturer device and three student devices. Distances were estimated by the tester rather than measured with laboratory instrumentation. Acoustic observations used very close, best-case quiet, and noisy conditions. BLE observations were grouped into short, medium, and longer ranges and included common indoor obstructions. Consequently, the field results are reported as successful observations and practical ranges, not as statistical detection probabilities.

Table 4.1
Android Devices Used in Physical Testing

Device	Role	Use during evaluation
Redmi 13C	Principal lecturer	Session creation, acoustic transmission, lecturer BLE advertising
POCO C71	Student	Acoustic/BLE scan and attendance submission
Samsung A05	Student	Acoustic/BLE scan and attendance submission
Vivo Y33	Student	Acoustic/BLE scan and attendance submission

The lecturer workflow was also tried on another available phone to confirm that session broadcasting was not restricted to the Redmi 13C. The Redmi remained the controlled lecturer reference for the observations presented here.

4.4 Automated Verification Results

The final code audit used focused tests instead of relying on a long-running static analyser. The backend suite created an isolated test database and exercised API behaviour. Flutter widget and unit tests rendered the principal role layouts and checked storage behaviour. The Android Gradle task compiled the Kotlin implementation offline.

Table 4.2
Automated Software Verification

Verification	Scope	Result
Django API tests	15 tests covering authorisation, sessions, proof validation, duplicate/replay logic, beacon rules, device uniqueness, privacy, and logout	15 passed
Django system check	Model, URL, application, and configuration checks	No issues reported
Migration consistency	Compare models with committed migrations	No pending migration changes
Python byte-code compilation	attendance and config packages	Completed without syntax error
Flutter tests	Authentication screen, student mobile layout, lecturer mobile layout, device-ID persistence	4 passed
Android Kotlin compilation	Foreground service, audio components, BLE advertiser, plugin integration	Build successful

The backend tests included a case in which two different students submitted the same classroom BLE nonce. Both submissions were accepted because the signal was valid for both authenticated students. A repeat by the same student was rejected. This result verifies that replay uniqueness is correctly student-scoped rather than globally blocking every classmate after the first submission.

Other tests confirmed that a student could not update a lecturer session, a non-owner could not manage another lecturer's session, a changed proof digest was rejected, a submitted identity had to match the signed-in student, unsupported proof was rejected, a compact acoustic token was accepted, a beacon had to match the room and RSSI rule, one installation

identifier could not be assigned to two student profiles, private legacy fields were not returned in profile responses, and a logout token could no longer access the API.

4.5 Acoustic Results

Acoustic transmission and decoding worked on the four-device test set, establishing that the implemented frame could be exchanged between ordinary phone speakers and microphones. Reception was strongly dependent on separation, noise, loudness, and placement.

Table 4.3
Observed Acoustic Performance

Condition	Observed separation	Finding
Normal close test	Approximately 1-30 cm	Decodes were obtained, with success affected by noise and device placement
Quiet room, maximum lecturer volume	Approximately 50 cm or slightly more	Furthest successful observation reported during the test period
Increased noise or less favourable placement	Beyond the reliable close range	Decode became inconsistent or failed

No success percentage is reported because the number of attempts per distance and noise condition was not fixed or logged. The result supports feasibility but does not support a claim of classroom-wide acoustic coverage.

The observed range is lower than that reported by Getreuer et al. (2018). This difference is technically plausible because their system used direct-sequence spread spectrum and a protocol designed for robust consumer-hardware communication, whereas the project uses a compact binary frequency-shift-keyed frame and relatively simple tone decisions. The difference also shows why a result from another protocol cannot be adopted as the expected range of this implementation.

The acoustic path remains useful as independent very-short-range copresence evidence. It is not the principal mechanism for a medium or large room. Increasing output

alone is not an adequate solution because greater amplitude may increase audible artefacts without overcoming microphone response or noise. Further improvement would require controlled work on coding gain, synchronisation, filtering, error correction, repeated capture, and device calibration.

4.6 Lecturer-Device BLE Results

Lecturer-device BLE was the strongest practical signal path in the evaluation. Each of the three student phones detected the lecturer advertisement in the short, medium, and longer distance groups used by the tester.

Table 4.4
Observed Lecturer BLE Performance

Distance group	Estimated range	Result on POCO C71, Samsung A05, and Vivo Y33
Short	1-5 m	Lecturer BLE detected
Medium	5-10 m	Lecturer BLE detected
Longer	Above 10 m	Lecturer BLE detected under the tested conditions

BLE also remained detectable during tests involving curtains or fabric, wooden objects, other room contents, and walls. These observations do not define a guaranteed maximum range. They show that the lecturer phone provided useful coverage beyond the current acoustic path and tolerated several non-line-of-sight conditions.

The result agrees with the literature in treating BLE as practical classroom technology (Azmi et al., 2018; Hayati & Nugraha, 2023; Noguchi et al., 2015; Puckdeevongs et al., 2020). It also reinforces the limitation described by Ramirez et al. (2021): detection and RSSI are affected by hardware and environment. A BLE signal observed through a wall is useful for coverage but creates adjacent-room ambiguity. The server's room, open-session, freshness, and RSSI checks are therefore necessary.

4.7 DX-CP27 Room-Beacon Results

The DX-CP27 Mini was configured and detected through its BLE frames. The project supports both iBeacon and Eddystone UID parsing, and the configured frame can be stored as a RegisteredBeacon assigned to a room. This verified that a commercial beacon could participate in the implemented scan and backend workflow without pairing with every student phone.

The lecturer-device advertisement was selected more often than the CP27 during combined tests, even when the beacon was physically close in some trials. The source-selection logic was subsequently revised to compare the strongest lecturer and beacon observations. If one exceeds the other by at least 10 dB, the stronger source is selected; if the readings are within 10 dB, both are retained.

Table 4.5
Room-Beacon Evaluation

Area	Result
CP27 frame visibility	Detected during BLE scanning
iBeacon/Eddystone parsing	Supported by the student scanner
Backend registration	Beacon identity, room, RSSI, and configuration stored
Session resolution	Open session resolved from the beacon's registered room
Combined source selection	10 dB comparison implemented; lecturer source had dominated earlier observations
Large-room and adjacent-room reliability	Not established by the completed small-scale tests

The fixed beacon is not a Bluetooth amplifier. It is an additional transmitter whose known identity can provide coverage at another point in the room. Its signal is static and does not satisfy the 60-second lecturer freshness rule. Instead, the backend requires a currently open session in the registered room and applies the beacon's minimum RSSI. This avoids accepting attendance simply because the beacon was powered before class.

4.8 Permission, Bluetooth State, and Hosting Results

Android permission state had a direct effect on BLE. Before the required location-related access was enabled on one test configuration, BLE appeared to have poor range. After permission was granted, detections extended to the distance groups reported in Table 4.4. This observation is consistent with the Android permission requirements for BLE scanning and advertising (Android Developers, n.d.).

The application now checks Bluetooth state before BLE scanning and returns a direct instruction when the adapter is off. Microphone, nearby-device, Bluetooth, and location permissions are requested at the point of use. The updated Bluetooth-off and background-broadcast behaviours were compiled successfully; they should remain part of the repeat physical regression checklist for each release APK.

The hosted backend removed the principal LAN limitation of earlier builds. Both test phones could communicate with the same stable URL without USB or a changing desktop IP. When the free Render instance had slept, its initial response took approximately one to two minutes. Subsequent requests were normal for the small-scale workflow. This delay is acceptable for a demonstration if the service is opened in advance, but it is not suitable evidence of production availability.

4.9 Functional and User-Experience Results

Registration, login, role routing, session creation, scan, submission, history, report retrieval, and CSV export operated during development tests. User-facing errors were revised so that connection failures, disabled Bluetooth, denied permissions, expired signals, duplicates, room mismatch, and server cold starts produce concise recovery guidance instead of socket traces or serializer dumps.

The lecturer Live and Reports layouts were restructured for class-size growth. Search is available across the relevant session and student fields. Student records are compact and

expand on demand. Flutter viewport tests did not report layout overflow for the student or lecturer portal at 390 x 844 logical pixels.

The report is session-specific. Changing the selected session changes the displayed records, course, lecturer, and room context. The CSV export is generated from the same selected session to avoid combining unrelated classes.

4.10 Attendance-Control Results

The small-scale evaluation and automated suite produced evidence for the following controls:

1. a student identity must match the authenticated profile;
2. the installation identifier survives logout;
3. one non-empty installation identifier cannot belong to two student profiles;
4. one student cannot create more than one attendance proof for a session;
5. two classmates can legitimately use the same broadcast nonce;
6. the same student cannot replay the same acoustic challenge or BLE nonce in the session;
7. a changed proof digest is rejected;
8. a registered beacon must match the selected session room and RSSI policy;
9. only the owning lecturer may manage or report on a session; and
10. logout revokes the server authentication token.

These results should not be interpreted as complete fraud prevention. Device binding does not detect who is physically carrying the registered phone, and SHA-256 without a protected secret does not make a modified client trustworthy. Real-time relay, rooted-device manipulation, collusion, and institutional device-recovery policy remain outside the completed evaluation.

4.11 Discussion

The results support a BLE-centred attendance design. Lecturer-device BLE provided useful detection on all three student phones at estimated ranges above 10 m, while the acoustic path was normally limited to 1-30 cm. Requiring both signals for every student would therefore cause unnecessary rejection. Treating acoustic, lecturer BLE, and registered room beacon as alternative valid paths is consistent with the observed performance.

The CP27 provides a practical infrastructure option rather than a replacement for the lecturer session. A fixed beacon can improve spatial coverage, but its static frame creates the same class of security concern identified by Kim et al. (2018). The project's response is contextual validation: a beacon must be registered to the room, an owned session must be open, the RSSI must be acceptable, and the student/device/duplicate checks must pass.

The four-device result is encouraging but not statistically generalisable. Hayati and Nugraha (2023) reported a device that failed to perceive a beacon in their evaluation, demonstrating that handset variation remains possible even when the present four devices behaved similarly. Larger tests should therefore retain device model and Android version in the test record.

The automated results strengthen the implementation claims that cannot be inferred from a visual demonstration. A successful scan animation does not prove that another student cannot reuse the nonce or that a non-owner cannot edit a session. Those claims are supported by server tests and database constraints. Conversely, automated tests cannot establish radio range or classroom usability; those claims are limited to the physical observations.

4.12 Limitations of the Evaluation

The evaluation has the following limitations:

1. only four Android devices were included;

2. the physical tests were repeated informally rather than with a fixed number of trials per condition;
3. distances were estimated rather than instrumented;
4. no calibrated sound-pressure, ambient-noise, or spectrum measurement was taken;
5. RSSI values were not logged systematically for every distance and phone orientation;
6. a complete class did not submit concurrently;
7. several adjacent rooms with simultaneous sessions and beacons were not field-tested;
8. lecturer foreground broadcasting requires repeat physical verification on the final installed APK across navigation, screen lock, and manufacturer battery settings;
9. the free hosted service was not load- or availability-tested; and
10. the evaluation did not include iOS, accessibility participants, formal penetration testing, or long-term operation.

These limitations constrain the conclusions but do not invalidate the verified functionality. They define the boundary between a working final-year prototype and an institution-ready attendance service.

4.13 Summary

The system was implemented and verified across mobile, native Android, backend, database, and hosted-deployment components. Fifteen backend tests and four Flutter tests passed, Django reported no system or migration issue, and the Kotlin implementation compiled successfully.

Physical observations showed that the acoustic path worked at very close range, normally about 1-30 cm and at slightly above 50 cm in the best reported quiet condition. Lecturer-device BLE was detected by the three student phones at estimated short, medium, and above-10-metre ranges and through several ordinary indoor obstacles. The CP27 was

detected and integrated as a registered room beacon, but its placement and multi-room performance require more formal evaluation.

The evidence supports BLE as the principal proximity mechanism, acoustic proof as supplementary close-range evidence, and the fixed beacon as optional room infrastructure governed by server-side context. Chapter Five presents the conclusions and recommendations that follow from these findings.

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 Summary of the Study

This study addressed the need for a faster and more accountable classroom attendance process in higher education. Manual roll call and paper registers consume lecture time, require later compilation, and can permit proxy attendance. Basic digital forms improve storage but do not establish that the authenticated device observed evidence associated with the classroom.

The project designed and implemented a smart attendance system that combines an Android application, acoustic communication, Bluetooth Low Energy proximity evidence, and a hosted Django validation service. The lecturer creates a course session, selects a room, opens attendance, and starts rotating acoustic and BLE signals. A student scans for a valid signal and submits a proof. A fixed DX-CP27 beacon may also provide registered room evidence.

The server determines whether attendance is accepted. It checks the authenticated identity, session owner and state, attendance window, signal format, session match, proof time, cryptographic digest, room-beacon registration, RSSI rule, device association, duplicate record, and student-scoped replay record. This arrangement prevents the mobile interface from declaring attendance without server approval.

The project followed an iterative Agile process. Acoustic, lecturer BLE, fixed-beacon support, device binding, reporting, hosted deployment, permission guidance, and interface refinements were added and revised through physical-device use.

Verification combined 15 backend API tests, four Flutter tests, Django checks, native Android compilation, and small-scale physical testing with a Redmi 13C, POCO C71, Samsung A05, and Vivo Y33. The physical tests covered acoustic and BLE reception,

estimated distance groups, common obstacles, Android permissions, CP27 detection, hosted connectivity, proof submission, and reporting.

5.2 Achievement of the Objectives

The first objective was to design a mobile attendance system through which a lecturer could create sessions, select rooms, open attendance, and manage records. This objective was achieved. The lecturer portal supports owned-session management, live attendance, search, session-specific reporting, permanent deletion, and CSV export.

The second objective was to implement acoustic and BLE proximity verification. Both channels were implemented on Android. The lecturer phone transmits compact acoustic frames and rotating BLE service data. The student app captures both channels concurrently and also parses registered iBeacon and Eddystone UID frames. The acoustic mechanism is functional at close range; lecturer BLE provides the principal measured coverage.

The third objective was to develop server-side validation for integrity, freshness, room, RSSI, duplicate, replay, and device controls. The Django API and database implement these rules. Automated tests verified the critical authorisation, identity, digest, duplicate, replay, beacon, and device-uniqueness cases.

The fourth objective was to provide role-appropriate interfaces. Lecturer and student portals, permission guidance, scan feedback, history, reports, profile/device information, search, expandable student rows, and CSV export were implemented. Mobile viewport tests completed without layout overflow for the tested student and lecturer screens.

The fifth objective was to evaluate the system on real Android devices under different conditions. A small-scale four-device evaluation was completed. It established functional acoustic transfer at very close range and lecturer BLE detection across the estimated distance groups. The objective was achieved within the declared prototype scope, but the evaluation is not broad enough to support institution-wide performance claims.

5.3 Major Findings

The first finding is that BLE is the most practical proximity channel in the present implementation. Each of the three student phones detected the lecturer advertisement at estimated ranges of 1-5 m, 5-10 m, and above 10 m under the tested conditions. BLE also remained detectable through several ordinary indoor obstacles.

The second finding is that the acoustic design works but does not provide classroom-wide range. Reception was generally observed between approximately 1 and 30 cm. A distance slightly above 50 cm was reached in the best reported quiet and high-volume condition. Noise, placement, loudness, speaker response, and microphone sensitivity affected decoding. Acoustic proof is consequently retained as supplementary close-range evidence.

The third finding is that a commercial fixed beacon can be integrated without becoming the session authority. The CP27 frame was detected and registered to a room. Its static identity is accepted only when a session is open in that room and the RSSI rule is satisfied. This avoids treating a powered beacon as permanent attendance permission.

The fourth finding is that permission handling can be mistaken for radio performance. BLE range improved materially after the required Android location-related permission was enabled. Bluetooth state and permission guidance are therefore part of the operational design, not merely interface details.

The fifth finding is that fraud controls must be scoped correctly. A classroom nonce is intentionally shared among students, so global nonce uniqueness would reject legitimate classmates. Student-scoped replay rules allow independent submissions while preventing the same student from reusing the signal. Device binding prevents one installation identifier from being assigned to several student accounts but cannot prove who is physically holding the registered phone.

The sixth finding is that hosting improves repeatability of the installed application. The Render deployment removed dependence on USB and a changing local IP address. Its free-tier cold start of approximately one to two minutes remains an availability limitation.

5.4 Conclusion

The study concludes that a BLE-centred, server-validated attendance system is feasible on the tested Android devices. The completed prototype supports the full path from lecturer session creation to student signal capture, server validation, session-specific reporting, and export. It offers stronger evidence than a static online form because acceptance requires an authenticated student, an open lecturer-owned session, and an authorised physical-signal path.

Lecturer-device BLE currently provides the best balance of coverage, speed, and reliance on equipment already available in the phones. A registered room beacon can extend or stabilise coverage, particularly when its location and RSSI policy are configured for the room. Neither signal should be treated as exact positioning. BLE propagation beyond a wall and the possibility of imitation or relay require contextual server controls.

The acoustic channel demonstrates a second telecommunication path and provides close-range copresence evidence. Its measured range is too short and noise-sensitive for use as the sole mechanism in a normal medium or large class. Retaining it as an alternative rather than a mandatory second signal prevents the weakest channel from reducing overall availability.

The system reduces several common forms of misuse through identity matching, installation binding, signal rotation, digest comparison, room registration, duplicate prevention, replay records, and lecturer ownership. It does not eliminate collusion, live relay, or use of another person's registered phone. The prototype should therefore be described as fraud-aware rather than fraud-proof.

Within the stated four-device and small-scale scope, the project achieved its aim. It is suitable for academic demonstration and structured pilot testing. Wider deployment would require quantitative field trials, production hosting, operational policy, stronger device assurance where available, and institutional review of privacy and exception handling.

5.5 Contributions of the Study

The main technical contribution is the integration of three alternative proximity observations within one attendance-validation workflow:

1. a compact rotating acoustic token;
2. a rotating lecturer-device BLE nonce; and
3. a static registered room-beacon identity constrained by session and room context.

The study also contributes a practical BLE source-selection method. The scanner compares lecturer and beacon RSSI observations, selects one source when it is at least 10 dB stronger, and retains both when their values are within the margin. This makes beacon use responsive to the observed signal rather than applying an unconditional lecturer-first preference.

A further contribution is the student-scoped replay model. It recognises that a broadcast is shared public classroom evidence and that many authenticated students must be able to submit the same current nonce. Replay prevention is applied to one student and session rather than incorrectly consuming the signal globally.

The room-aware beacon design contributes an operational method for nearby classrooms. A beacon has a registered room and cannot resolve attendance unless one unexpired session is open in that room. Only one session may be open for the room at a time. These rules reduce ambiguity without requiring the beacon itself to receive a new lecture schedule.

From a Telecommunication Science perspective, the implementation demonstrates applied acoustic modulation, frequency-selective filtering, BLE advertising, RSSI interpretation, indoor propagation effects, mobile-network communication, and layered validation. The study also documents the difference between theoretical feasibility and measured performance on consumer hardware.

5.6 Limitations

The study is limited by the size and formality of its evaluation. Four Android phones cannot represent the full range of speakers, microphones, Bluetooth radios, antenna layouts, firmware, and power-management policies. Physical distances were estimated, trial counts were not fixed, and noise and sound pressure were not measured.

The acoustic frame is not sufficiently robust for room-wide use. Its two-tone design, 12 ms bit windows, and simple checksum provide a functional prototype but less coding gain and error correction than established ultrasonic communication protocols.

BLE RSSI is an approximate and device-dependent observation. The configured minimum RSSI can reject very weak signals but cannot draw a precise geometric boundary around a classroom. Multiple rooms, crowds, interference, phone orientation, and real-time relay require further evaluation.

The DX-CP27 was integrated but not tested across a full multi-beacon deployment. The current findings do not establish the number or placement of beacons required for a large lecture theatre.

The installation identifier is generated and stored by the application. It is useful for continuity and account/device policy but is not hardware-backed attestation. A legitimate device replacement also requires an administrative reset process.

The free hosted service provides no institutional availability guarantee. The implementation has not undergone formal penetration testing, high-volume load testing,

disaster-recovery testing, accessibility evaluation, iOS implementation, or long-term operational assessment.

5.7 Recommendations

BLE should remain the principal proximity mechanism for the present version. Lecturer-device advertising avoids immediate infrastructure cost, while registered room beacons should be introduced only where measured room coverage justifies them.

Each deployed beacon should have an inventory record containing its room, format, UUID or namespace, major/minor or instance value, transmit power, advertising interval, mounting position, installation date, and minimum RSSI policy. Beacon values should not be treated as confidential credentials.

The acoustic path should remain optional while its receiver is improved. Any redesign should be compared through fixed trial counts at measured distances and noise levels. More robust synchronisation, forward-error correction, coding gain, and device calibration are preferable to simply increasing volume.

The next field evaluation should record device model, Android version, room dimensions, phone orientation, obstruction, distance, RSSI, signal source, scan duration, submission outcome, and elapsed time. A fixed number of attempts per condition would support detection rate, false rejection, and confidence intervals.

Adjacent-room tests should open simultaneous lecturer sessions and operate multiple registered beacons. A successful evaluation should demonstrate that a student in one room does not resolve or submit against the other room merely because its BLE signal is detectable.

Device changes should require a documented recovery process. An administrator should verify the student's identity and reason, reset the old binding, and retain an audit entry. Automatic unrestricted re-binding would defeat the one-installation rule.

Institutional deployment should use a paid, monitored hosting service or managed university infrastructure with PostgreSQL backups, HTTPS, secret rotation, health monitoring, audit logging, rate limiting, availability targets, and tested recovery procedures.

Attendance records should remain subject to human review. A technical rejection caused by a dead phone, denied permission, inaccessible hardware, or signal interference should have a controlled exception process rather than automatically becoming an academic penalty.

5.8 Suggestions for Further Work

Further acoustic research may compare binary frequency-shift keying with chirp spread spectrum or direct-sequence methods, investigate adaptive thresholds, and add stronger error detection or correction. Testing should include spectral measurements and user audibility at the selected frequencies.

BLE work should evaluate calibrated RSSI policies for different rooms and handset groups. Multiple observations over a scan period may be filtered to reduce a single transient reading. Beacon-only, lecturer-only, and combined-source trials should be reported separately.

An offline-first extension could queue an already captured proof during a brief network interruption. Such a queue would need protected local storage, server-signed session material, strict expiry, and replay-safe synchronisation; otherwise, offline support could weaken freshness.

Stronger Android device assurance may be investigated through platform attestation and signed server challenges. This would raise implementation complexity and still would not prove that the registered student is carrying the phone.

An iOS client may be developed after the Android workflow is stable. Background BLE advertising, audio capture, and operating-system restrictions would need an independent design and evaluation rather than a direct assumption of parity.

Finally, a controlled pilot with a complete class should measure simultaneous submissions, lecturer workload, student completion time, error recovery, report accuracy, and perceived usability over several sessions. Those data would determine whether the prototype is suitable for departmental adoption.

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